

A Modern Biostatistical Re-analysis of the NCCTG Adjuvant Colon Cancer Randomized Trial

Frequentist, Bayesian, and Machine Learning Perspectives

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Abstract

Background. Adjuvant chemotherapy with levamisole and 5-fluorouracil (5-FU) after resection of stage B/C colon carcinoma was established by the North Central Cancer Treatment Group (NCCTG) program in the late 1980s, and the resulting individual patient dataset has become one of the most widely used teaching resources in survival analysis. This project asks what a modern biostatistician can extract from the same trial today, when equipped with contemporary methods for missing data, sequential monitoring, simulation, machine learning, Bayesian inference, and heterogeneous treatment effects.

Methods. Among 929 patients randomized to observation ($n = 315$), levamisole ($n = 310$), or levamisole plus 5-FU ($n = 304$), a composite disease-free survival (DFS) endpoint (recurrence or death; 506 events) was reconstructed from the published event-type coding, and overall survival (452 deaths) served as the secondary endpoint. The re-analysis proceeds in six modules: (i) Kaplan–Meier estimation, log-rank tests, Cox models with multiple imputation ($m = 20$), competing-risk and restricted-mean-survival-time sensitivity analyses, proportional-hazards diagnostics, and bootstrap absolute benefit; (ii) a redesign of the trial as a three-look group-sequential design with O’Brien–Fleming efficacy spending and non-binding beta-spending futility boundaries, including a re-enactment of interim monitoring on the real event times; (iii) Monte Carlo simulation studies quantifying the efficiency of covariate adjustment, the operating characteristics of the sequential design, and confidence-interval coverage under non-proportional hazards; (iv) cross-validated survival prediction with five machine-learning models; (v) a Bayesian Weibull proportional-hazards analysis with skeptical, weakly-informative, and enthusiastic priors, sequential monitoring, and posterior predictive checks; and (vi) doubly robust estimation of average and conditional treatment effects on the five-year DFS event risk using jackknife pseudo-observations.

Results. Levamisole plus 5-FU reduced the hazard of the composite DFS endpoint by 38% (adjusted hazard ratio 0.621, 95% CI 0.496–0.776), with consistent evidence across unadjusted, imputation-adjusted, and Fine–Gray competing-risk analyses, and an absolute benefit of 16.8 percentage points at five years (59.2% vs. 42.4% DFS; number needed to treat = 6), and a gain of 0.63 disease-free years in restricted mean survival time within five years. Under the redesigned sequential plan, the first interim analysis after 118 events would have yielded a hazard ratio of 0.52 ($Z = 3.38$), crossing the efficacy boundary of 2.96; Bayesian monitoring with a skeptical prior would have reported $\Pr(\text{HR} < 1) = 0.998$ after only 25% of events. In simulation, covariate adjustment increased power from 72.6% to 80.9% for a moderate effect without inflating the type-I error, and directly illustrated the non-collapsibility of the hazard ratio. Random survival forests achieved the best validated discrimination (cross-validated $C = 0.646$), and doubly robust estimation located the largest absolute benefits in the highest-risk patients.

Conclusions. A single 1980s trial, re-analyzed with modern biostatistical machinery, supports a robust treatment effect and illustrates how contemporary design and monitoring standards would have accelerated its conclusion. The project demonstrates the complementary value of frequentist, Bayesian, and machine-learning perspectives within one coherent, fully scripted pipeline. All code, data, and outputs are reproducible end-to-end from the accompanying repository.

Keywords: randomized clinical trial; survival analysis; group-sequential design; multiple imputation; competing risks; Bayesian inference; machine learning; heterogeneous treatment effects.

1 Introduction

Colorectal cancer is one of the most common malignancies worldwide, and for patients undergoing potentially curative resection the question of whether systemic adjuvant chemotherapy should follow surgery remained open for much of the twentieth century. The North Central Cancer Treatment Group (NCCTG) program addressed this question directly: building on an earlier randomized comparison of observation against levamisole alone [1], the investigators added a third arm combining the immunomodulator levamisole with the antimetabolite 5-fluorouracil (5-FU) [2]. The combination reduced recurrence and mortality substantially and became a standard of care for resected stage B/C colon carcinoma. The individual patient data from this program were subsequently contributed to the `survival` package for R [3], where they now serve as one of the canonical datasets for teaching survival analysis; the version analysed here contains 929 patients from a single study.

Re-analyzing an old trial with new methods is more than an academic exercise, and five considerations motivated each analytical module of this project. First, endpoint definitions and analysis conventions have evolved: modern adjuvant colorectal trials define disease-free survival (DFS) as the time to recurrence *or* death, whereas the published coding of this dataset treats death without recurrence as censoring; a rigorous re-analysis must therefore address both views and, importantly, the competing-risks consequences of the difference. Second, missing-data handling, covariate-adjusted estimation, and absolute-benefit reporting have all been refined since the original publications, and regulatory guidance now actively encourages pre-specified covariate adjustment in the primary analysis of randomized trials [16, 15]. Third, the trial-design landscape has changed: group-sequential designs with alpha- and beta-spending functions are now the default in oncology, and it is natural to ask whether this trial—run with a fixed-sample mindset—would have stopped early under a modern monitoring plan [10, 9]. Fourth, machine-learning prediction models and Bayesian machinery offer perspectives on prognosis and evidence accumulation that complement the classical toolkit [21, 22]. Finally, the modern estimand framework and recent methods for heterogeneous treatment effects invite a re-examination of *for whom* the benefit is largest [19].

Accordingly, this project integrates all of these perspectives into a single coherent pipeline with six objectives:

1. **Re-estimate the treatment effects** on DFS and overall survival with covariate-adjusted Cox models fitted on multiply imputed data, explicit proportional-hazards diagnostics, competing-risk sensitivity analyses, and bootstrap and restricted-mean-survival-time absolute-benefit measures (Sections 3.1 and 4);
2. **Redesign the trial** as it would be designed today—a formal sample-size calculation and a three-look group-sequential design—and re-enact interim monitoring on the real event times (Sections 3.3 and 5);
3. **Quantify operating characteristics by simulation**, covering the efficiency of covariate adjustment, the type-I error and power of the sequential design, and confidence-interval coverage under non-proportional hazards (Sections 3.4 and 6);
4. **Build and honestly validate prognostic survival models** spanning classical regression, penalization, and ensembles of trees (Sections 3.5 and 7);
5. **Conduct a Bayesian re-analysis** with prior sensitivity, event-driven sequential monitoring, posterior predictive checks, and baseline-model criticism (Sections 3.6 and 7.2);
6. **Characterise heterogeneous treatment effects** on the absolute five-year risk using jack-knife pseudo-observations, cross-fitted G-computation, and doubly robust AIPW estimation (Sections 3.7 and 8).

The report is organized to mirror the repository structure: Section 2 describes the data and endpoint construction; Section 3 details all statistical methods; Sections 4–8 present results in five thematic blocks; Section 9 discusses findings, strengths, and limitations. All analyses are scripted in R and Python with fixed random seeds, and Section 3.8 records the exact software environment required for byte-level reproduction.

2 Data

2.1 Source and patients

The analysis dataset derives from the `colon` data frame distributed with the R `survival` package [3], obtained as a CSV file from the Rdatasets mirror and stored in the repository under `data/raw/`. The underlying trial was conducted by the NCCTG and reported by Laurie et al. [1] and Moertel et al. [2]. The file stores *two records per patient*: one for time-to-recurrence and one for time-to-death, with baseline covariates duplicated across the pair. After verifying covariate consistency within each patient (all 929 patients passed), the two records were pivoted into a single patient-level analysis file of 929 rows.

Patients were randomized 1:1:1 within a single study to observation (**Obs**, $n = 315$), levamisole alone (**Lev**, $n = 310$), or levamisole plus 5-FU (**Lev+5FU**, $n = 304$) following surgical resection. Baseline covariates available for adjustment and prediction comprise: age (mean 59.8 years), sex (47.9% female), obstruction of the colon by tumour (19.4%), perforation (2.9%), adherence to nearby organs (14.5%), number of positive lymph nodes (median 2, IQR 1–5), tumour differentiation (well/moderate/poor), extent of local spread (submucosa through contiguous structures), and the surgery-to-registration interval category (short/long).

2.2 Endpoint construction

The published coding defines the `etype = 1` record as time to tumour recurrence, with death before recurrence treated as censoring. In the present dataset this coding produces 468 recurrences and, notably, 38 deaths that occur without a preceding recurrence flag at the same visit date. Because contemporary adjuvant colorectal trials define DFS as recurrence *or* death from any cause, the data-curation module constructed three complementary endpoint views:

1. **Composite DFS** (primary endpoint): time to first of recurrence or death, whichever occurs first; 506 events (54.5%).
2. **Overall survival** (secondary endpoint): time to death from any cause; 452 events (48.7%).
3. **Recurrence with competing events** (sensitivity analyses): cause-coded status with 468 recurrences (cause 1), 38 competing deaths (cause 2), and 423 censored observations, supporting Aalen–Johansen estimation and Fine–Gray subdistribution modelling.

Cross-endpoint consistency checks confirmed that the recurrence time never exceeds the death time and that all 38 competing deaths occur exactly at their death dates. Follow-up is substantial: the median follow-up among censored patients, estimated by reverse Kaplan–Meier, is 6.4 years, and the maximum observed time is 9.1 years.

2.3 Missing data and imputation

Two adjustment covariates contain missing values: the number of positive lymph nodes (18 patients, 1.9%) and tumour differentiation (23 patients, 2.5%); 95.6% of patients are complete cases. Following White and Royston’s recommendation for survival outcomes [5], missing covariates were imputed by fully conditional specification (chained equations) using the `mice` package [6]

with $m = 20$ imputations, including the Nelson–Aalen cumulative hazard estimate and the event indicator of the relevant endpoint in each imputation model so that the imputation is compatible with the subsequent Cox analysis. A plausibility check comparing observed and imputed distributions showed modest and clinically reasonable differences (mean imputed nodes 3.3 vs. observed 3.7), and pooled estimates were contrasted against complete-case results throughout (Section 4).

2.4 Baseline balance

Randomization produced well-balanced arms. Across eleven baseline variables, the only nominal imbalance was sex (53.6% female in **Lev+5FU** vs. 47.3% in **Obs**; three-arm global exact test $p = 0.027$, pairwise $p = 0.13$), which is unremarkable in the context of eleven simultaneous tests and does not survive multiplicity correction. Pairwise standardized mean differences (Figure 1) never exceeded $|0.216|$, and the overall picture is consistent with adequate randomization concealment.

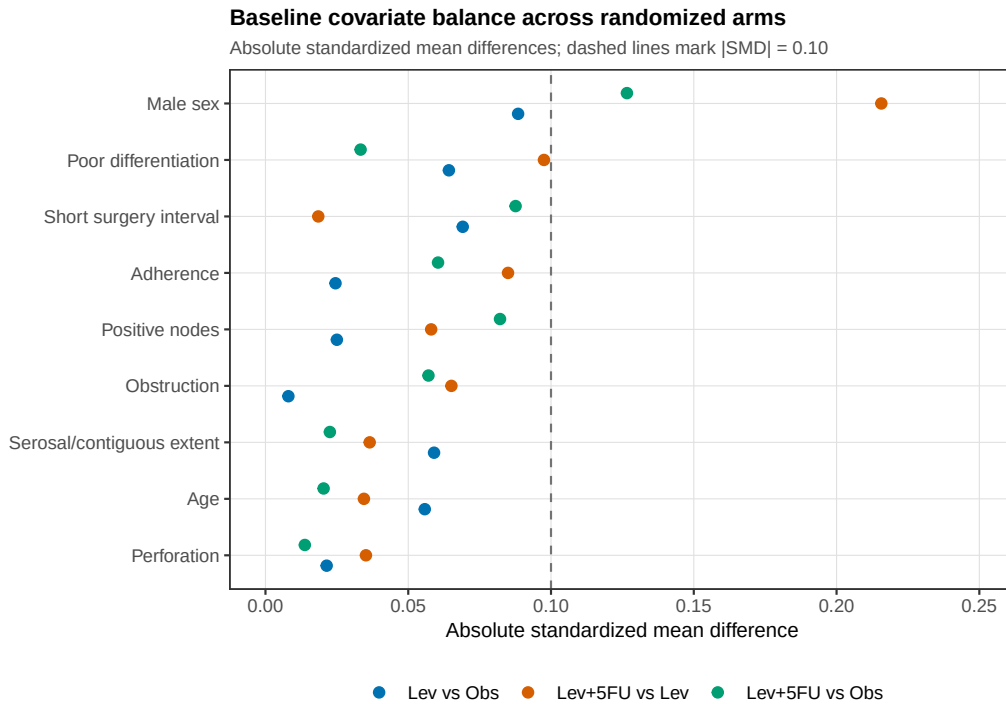


Figure 1: Baseline covariate balance across randomized arms, measured by absolute standardized mean differences (SMD). Dashed lines mark the conventional $|SMD| = 0.10$ threshold. All pairwise comparisons between arms are shown.

3 Methods

3.1 Frequentist efficacy analysis

Disease-free survival and overall survival were summarized by Kaplan–Meier curves with point-wise 95% confidence intervals (log–log transformation) and compared with log-rank tests, both overall (2 degrees of freedom) and pairwise between arms adjusted for multiplicity by closure of the testing family. Treatment effects were estimated with Cox proportional-hazards models fitted by Efron’s method for ties. Three specifications are reported for the primary endpoint: an unadjusted model; a covariate-adjusted complete-case model; and the pre-specified covariate-adjusted model fitted on each of the $m = 20$ imputed datasets and pooled by Rubin’s rules [4], with degrees of freedom adjusted for the fraction of missing information. The adjustment set

comprised treatment arm, age, sex, obstruction, perforation, adherence, number of positive nodes, differentiation, extent of local spread, and surgery-to-registration interval. An analogous pooled model was fitted for overall survival.

The proportional-hazards assumption was examined with scaled Schoenfeld residuals [7], both graphically (loess smooths against time) and with the global chi-square test. Two pre-specified sensitivity analyses address potential violations and endpoint-definition concerns: first, flexible functional forms for age and nodes via restricted penalized splines (4 degrees of freedom each), compared with the linear model by likelihood-ratio test; second, competing-risk analyses on the recurrence endpoint, contrasting the Aalen–Johansen cumulative incidence function with the naive one-minus-Kaplan–Meier estimate, and fitting Fine–Gray subdistribution-hazard models [8] adjusted for the same covariate set.

Because hazard ratios are not directly interpretable as absolute benefit, five-year Kaplan–Meier survival probabilities with 95% confidence intervals were computed by arm for both endpoints, along with between-arm risk differences and numbers needed to treat (NNT) whose 95% confidence intervals were obtained by nonparametric bootstrap with 2,000 resamples (resampling patients, recomputing the Kaplan–Meier estimates each time).

Because two covariates violated proportional hazards (and because the hazard ratio is a non-collapsible summary), the primary comparison was additionally re-expressed through the restricted mean survival time (RMST) [25, 26]: the area under the Kaplan–Meier survival curve up to a horizon τ , estimated at $\tau = 3, \dots, 8$ years with percentile bootstrap confidence intervals (2,000 resamples), and summarized both as a between-arm difference (disease-free years gained) and as a ratio. Unlike the hazard ratio, the RMST difference is collapsible and requires no proportional-hazards assumption. A covariate-adjusted RMST difference at $\tau = 5$ years was obtained by regressing leave-one-out jackknife pseudo-observations of RMST(5) [17, 18] on treatment and the standardized adjustment set, with bootstrap standard errors; because the estimand is collapsible, the adjusted and unadjusted analyses target the same quantity, providing a direct empirical contrast with the non-collapsibility of the hazard ratio quantified in Simulation Study A. RMST differences were also computed within the obstruction stratum, the covariate with the strongest proportional-hazards violation.

3.2 Subgroup and interaction analyses

Seven clinically motivated subgroups were pre-specified for the primary comparison of levamisole plus 5-FU against observation: age (< 60 vs. ≥ 60 years), sex, nodal burden (< 4 vs. ≥ 4 positive nodes), obstruction, local extent (submucosa/muscle vs. serosa/contiguous), differentiation (well/moderate vs. poor), and surgery-to-registration interval. Within each subgroup level, unadjusted Cox models estimated subgroup-specific hazard ratios; effect modification was tested by Wald tests of treatment-by-subgroup interaction terms in models adjusted for the nodal count. Results are displayed as a forest plot with interaction p -values. Given seven interactions tested, only p -values below a Bonferroni-adjusted threshold of $0.05/7 \approx 0.0071$ would be interpreted as evidence of modification; this multiplicity-aware reading is emphasized in Section 5.

3.3 Trial redesign and re-enacted monitoring

To redesign the trial under modern standards, planning assumptions were taken from the re-analysis itself. The control arm’s five-year DFS of 42.4% and median DFS of 2.96 years imply an exponential baseline hazard $\lambda_0 = \log 2/2.96 = 0.234$ per year under the (approximate) constant-hazard reading; the clinically meaningful target was set at a hazard ratio of 0.65, close to the effect actually observed. A fixed one-sided superiority design at $\alpha = 0.025$ with 90% power, 1:1 allocation, three years of uniform accrual and a minimum of five years of additional follow-up,

requires 226 DFS events and approximately 322 patients (Schoenfeld’s formula [13], verified with `gsDesign::nSurv` [12]).

The modern redesign is a three-look group-sequential trial with analyses after 50% and 75% of the planned events and at completion. Efficacy boundaries use O’Brien–Fleming alpha-spending [9, 10]; futility boundaries use a non-binding Hwang–Shih–DeCani beta-spending with shape parameter -4 [11], which yields aggressive early stopping for futility under the null while allowing the sponsor to disregard the futility boundary without inflating the type-I error. The resulting maximum sample size is 235 events (approximately 336 patients), a 4.1% inflation over the fixed design—the standard price of interim looks.

Two complementary analyses probe the design’s behaviour. First, a power curve displays the required events and achieved power as a function of the target hazard ratio. Second, and more unusually, the monitoring plan was *re-enacted on the real trial data*: because the public dataset contains no calendar dates, an information-time snapshot of the interim stage was reconstructed by administratively censoring every patient’s follow-up at the time (years since randomization) at which the 118th DFS event of the actual trial was observed, refitting the Cox model on the censored-at-interim snapshot, and comparing the resulting Z -statistic with the design boundaries. Conditional power for the final analysis was computed under three assumptions for the remaining drift (null effect, design alternative, and the observed interim trend), with the sign convention that positive Z favours the experimental arm.

3.4 Simulation studies

Three Monte Carlo studies quantify the operating characteristics behind the design and analysis choices. The data-generating mechanism was calibrated to the trial itself: covariates were drawn by bootstrap resampling of the complete-case patients, the prognostic index used hazard-ratio coefficients from the adjusted Cox model fit of Section 4, event times were exponential with hazard $\lambda_0 \exp(\text{lp}_i + \log(\text{HR}) \cdot \text{treat}_i)$, accrual times were uniform on three years, and administrative censoring occurred at eight years.

Study A (covariate-adjustment efficiency) simulated 1,000 trials per scenario for a two-arm comparison with $n = 322$ at hazard ratios of 1.0 (type-I error) and 0.72 (power), crossing three levels of prognostic strength obtained by scaling the calibrated covariate coefficients by 0, 0.5, and 1 (the corresponding standard deviations of the prognostic index are 0, 0.208, and 0.416 on the log-hazard scale). Each replicate fitted both the unadjusted Cox model and the fully-adjusted model, recording estimates, model-based standard errors, empirical standard deviations, and rejection frequencies, along with the log-rank statistic. Because the hazard ratio is non-collapsible [14], the unadjusted model targets the marginal hazard ratio, which is attenuated toward the null relative to the conditional (adjusted) estimand; the simulation quantifies both the power gain from adjustment and this attenuation explicitly.

Study B (group-sequential operating characteristics) simulated 3,000 trials per scenario under the design of Section 3.3 ($n = 335$, event triggers at 118/177/235). Interim snapshots were constructed by cutting each simulated trial at the calendar time of the planned event count; monitoring decisions followed the design boundaries, with futility either adhered to or ignored (the non-binding option), giving three scenarios: alternative with adherence, null with adherence, and null with futility ignored. The study reports stopping probabilities by analysis and reason, expected event counts, and the realized type-I error and power, for direct comparison with the design-theoretic values from `gsDesign`.

Study C (interval estimation under non-proportional hazards) simulated 500 trials of $n = 335$ in which the treatment effect was deliberately time-varying: hazard ratio 0.45 during the first year and 0.80 thereafter, a piecewise-constant pattern implemented through the memoryless property of the exponential distribution. Each replicate fitted an unadjusted Cox

model and recorded the Wald 95% confidence interval for the log hazard ratio, together with a 100-resample percentile bootstrap interval. Coverage was evaluated against the simulation estimand (the mean of the 500 point estimates, i.e. the quantity the estimator actually targets under misspecification), which is the honest comparison for studying interval reliability.

3.5 Machine-learning prediction

Prognostic modelling treats the composite DFS endpoint as a prediction problem, *deliberately excluding the randomized treatment indicator from all predictors* so that models quantify baseline risk only—a clean separation between the prediction task of this section and the causal tasks of Sections 3.6 and 3.7. Five models were compared on the 888 complete cases: (1) a clinical Cox model with the same main effects as the efficacy analysis; (2) an elastic-net penalized Cox model (ℓ_1 ratio 0.5, 50 alphas along the regularization path); (3) a random survival forest (400 trees, minimum leaf size 8, $\sqrt{p}$ features per split); (4) gradient boosting with the Cox partial-likelihood loss (250 trees, learning rate 0.05, minimum leaf size 8); and (5) XGBoost with the Cox objective (300 rounds, depth 3, 80% feature and row subsampling).

Honest validation used 5×5 -fold stratified cross-validation (five repeats of five folds, stratified on the event indicator; 25 training–test partitions per model). Discrimination was measured by Harrell’s concordance index and by the inverse-probability-of-censoring weighted (IPCW) AUC at five years; overall prediction error by the integrated Brier score over years 1–5, using the Kaplan–Meier censoring distribution for IPCW weighting [20]. In-sample (apparent) concordance is reported alongside the cross-validated values to expose the optimism gap. The best model by cross-validated concordance was then examined in depth: calibration by comparing mean predicted five-year risks with Kaplan–Meier-observed risks within deciles of predicted risk, a five-year time-ROC curve, and permutation importance (20 shuffles per covariate on a held-out 25% test split).

3.6 Bayesian re-analysis

The Bayesian analysis models the two-arm comparison (levamisole plus 5-FU vs. observation; $n = 607$ complete cases; 317 DFS events) with a Weibull proportional-hazards regression fitted by Hamiltonian Monte Carlo (No-U-Turn Sampler) in PyMC [23], with right censoring handled by the `Censored` likelihood wrapper. Writing the cumulative hazard as

$$H_i(t \mid x_i) = \left(\frac{t}{\beta_0 \exp(-\eta_i/\alpha)} \right)^\alpha, \quad \eta_i = \beta_{\text{tr}} \text{treat}_i + x_i^\top \gamma, \quad (1)$$

the shape parameter α permits increasing ($\alpha > 1$) or decreasing ($\alpha < 1$) baseline hazards, and $\exp(\beta_{\text{tr}})$ is the treatment hazard ratio. The covariate vector x_i contains age (per decade), $\log(1+\text{nodes})$, female sex, obstruction, advanced extent, and long surgery-to-registration interval; priors for covariate effects were weakly informative normals.

The centrepiece is a prior sensitivity analysis for the treatment effect on the log-hazard-ratio scale, comparing three specifications that span the spectrum of prior conviction: a *skeptical* prior $\mathcal{N}(0, 0.421^2)$, chosen so that only 5% of the prior mass lies below a hazard ratio of 0.5; a *weakly-informative* $\mathcal{N}(0, 1)$; and an *enthusiastic* prior $\mathcal{N}(\log 0.65, 0.421^2)$ centred at the design alternative. Posterior summaries (means, 95% credible intervals, and the probabilities $\Pr(\text{HR} < 1)$ and $\Pr(\text{HR} < 0.8)$) are contrasted with each other and with the frequentist estimate.

Bayesian sequential monitoring refits the model under the skeptical prior at four event-driven looks (25%, 50%, 75%, and 100% of the observed DFS events), each time censoring the data at the time of the corresponding event, exactly paralleling the frequentist re-enactment of Section 3.3 so that the two monitoring philosophies can be compared directly on the same evidence stream. Model criticism comprises (i) a posterior predictive check replicating Kaplan–Meier

curves for the control arm from posterior draws, and (ii) approximate leave-one-out cross-validation [24] comparing the Weibull baseline against an exponential special case ($\alpha \equiv 1$).

3.7 Heterogeneous treatment effects

The final module estimates average and conditional treatment effects on the *absolute* five-year DFS event risk for the two-arm comparison. Jackknife pseudo-observations [17, 18] convert the censored survival outcome into a continuous, regression-ready outcome: for patient i ,

$$\theta_i = n \hat{\theta} - (n - 1) \hat{\theta}_{(-i)}, \quad (2)$$

where $\hat{\theta}$ is the Kaplan–Meier five-year event probability in the full sample and $\hat{\theta}_{(-i)}$ the same quantity with patient i removed. On the modelling subset ($n = 594$ complete cases), the population-average treatment effect was estimated three ways: the unadjusted Kaplan–Meier risk difference with bootstrap confidence intervals; cross-fitted G-computation [19] predicting both arms’ pseudo-observations with LASSO outcome models fitted in the training folds; and a cross-fitted augmented inverse-probability weighted (AIPW) estimator

$$\hat{\psi}_{\text{AIPW}} = \frac{1}{n} \sum_{i=1}^n \left\{ \hat{\mu}_1(X_i) - \hat{\mu}_0(X_i) + \frac{T_i (\theta_i - \hat{\mu}_1(X_i))}{e} - \frac{(1 - T_i) (\theta_i - \hat{\mu}_0(X_i))}{1 - e} \right\}, \quad (3)$$

with influence-function standard errors, where $\hat{\mu}_a(\cdot)$ are the cross-fitted arm-specific outcome models and e is the (known by design, empirically estimated) propensity $P(T = 1) = 0.487$. Conditional effects were explored with a cross-fitted T-learner (LASSO within each arm), yielding patient-level predicted benefits $\hat{\mu}_0(X_i) - \hat{\mu}_1(X_i)$ whose distribution, dependence on baseline risk, and quartile-level survival curves are examined—with the explicit caveat that such exploratory heterogeneity estimation in a single trial of this size is hypothesis-generating only.

3.8 Software and reproducibility

All analyses were scripted in R 4.5.3 (packages `survival`, `mice`, `ggplot2`, `gsDesign`, and the `tidyverse` stack) and Python 3.12 (`scikit-survival`, `lifelines`, `xgboost`, `PyMC`, `ArviZ`); exact versions are recorded in `environment/` within the repository. A single global seed (20260916) plus module-specific seeds govern every stochastic step, from imputation to posterior sampling. The repository executes end-to-end via `make all`, regenerating every figure and table in this report in approximately 20–25 minutes on a single core. The pipeline has no hidden manual steps: data curation, quality-control assertions (record counts, covariate consistency, endpoint logic), analyses, and outputs are all code-reviewed and deterministic.

4 Results I: Treatment Effects

4.1 Kaplan–Meier analysis

Figures 2 and 3 display Kaplan–Meier estimates for the composite DFS endpoint and for overall survival, each with the numbers at risk. The three curves separate early and the separation widens over the first three years, driven almost entirely by the levamisole-plus-5-FU arm. The overall log-rank test is highly significant for DFS ($\chi^2 = 21.0$ on 2 degrees of freedom, $p < 0.001$); pairwise comparisons show that levamisole alone is indistinguishable from observation (HR 0.968, $p = 0.76$), whereas levamisole plus 5-FU is superior to both observation (HR 0.623, $p < 0.001$) and to levamisole alone ($p < 0.001$). The pattern for overall survival is similar but attenuated (adjusted HR 0.688, 95% CI 0.544–0.871), consistent with post-recognition salvage therapies diluting survival differences.

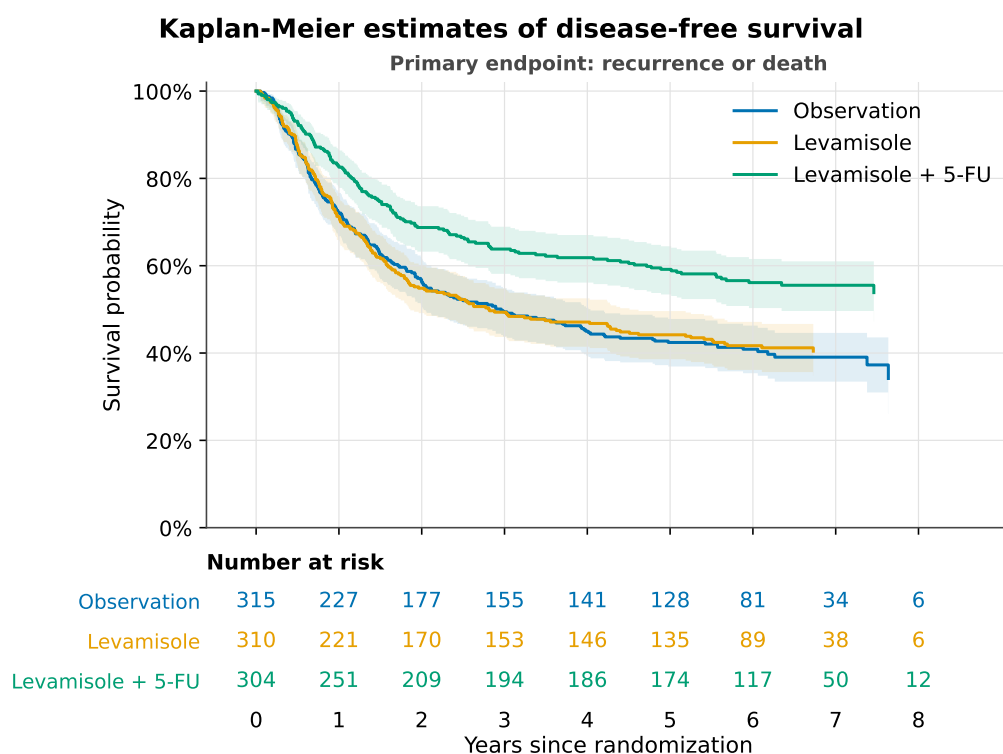


Figure 2: Kaplan–Meier estimates of the composite disease-free survival endpoint (primary) with pointwise 95% confidence bands and numbers at risk.

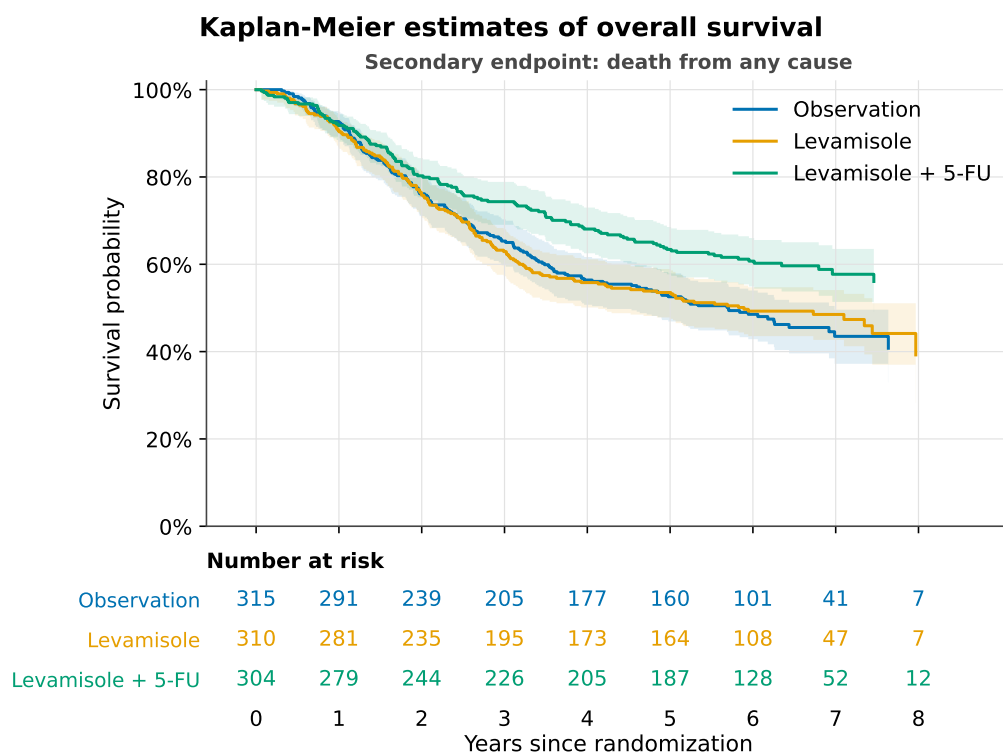


Figure 3: Kaplan–Meier estimates of overall survival (secondary endpoint) with pointwise 95% confidence bands and numbers at risk.

4.2 Cox models

Table 1 assembles the treatment-effect estimates across specifications. Adjustment for the ten baseline covariates, pooled over 20 multiple imputations by Rubin’s rules, moves the primary contrast only marginally (unadjusted HR 0.623 vs. adjusted 0.621), as expected under randomization with balanced arms; the complete-case analysis (not shown in full) agrees closely. The adjusted covariate effects themselves (Figure 4) are clinically coherent: each additional positive lymph node multiplies the hazard by 1.08 (95% CI 1.06–1.10), spread to contiguous structures triples the hazard relative to submucosal involvement (HR 3.19, 95% CI 1.32–7.75), obstruction raises it by 25%, and a long surgery-to-registration interval is protective in this cohort (HR 0.76, 95% CI 0.62–0.92)—the latter reflecting the dataset’s coding of a longer recovery interval before registration among patients who recovered well enough to be registered.

Table 1: Treatment-effect estimates for the composite DFS endpoint across analysis specifications. The Fine–Gray model targets the recurrence endpoint with death without recurrence as a competing event.

Analysis	Contrast	HR	95% CI	<i>p</i>
Unadjusted Cox	Lev vs. Obs	0.968	0.790–1.187	0.757
Unadjusted Cox	Lev+5FU vs. Obs	0.623	0.499–0.777	< 0.001
Adjusted Cox, MI ($m = 20$)	Lev vs. Obs	0.947	0.770–1.165	0.607
Adjusted Cox, MI ($m = 20$)	Lev+5FU vs. Obs	0.621	0.496–0.776	< 0.001
Adjusted Cox, MI (OS endpoint)	Lev+5FU vs. Obs	0.688	0.544–0.871	0.002
Fine–Gray subdistribution	Lev+5FU vs. Obs	0.596	0.469–0.756	< 0.001

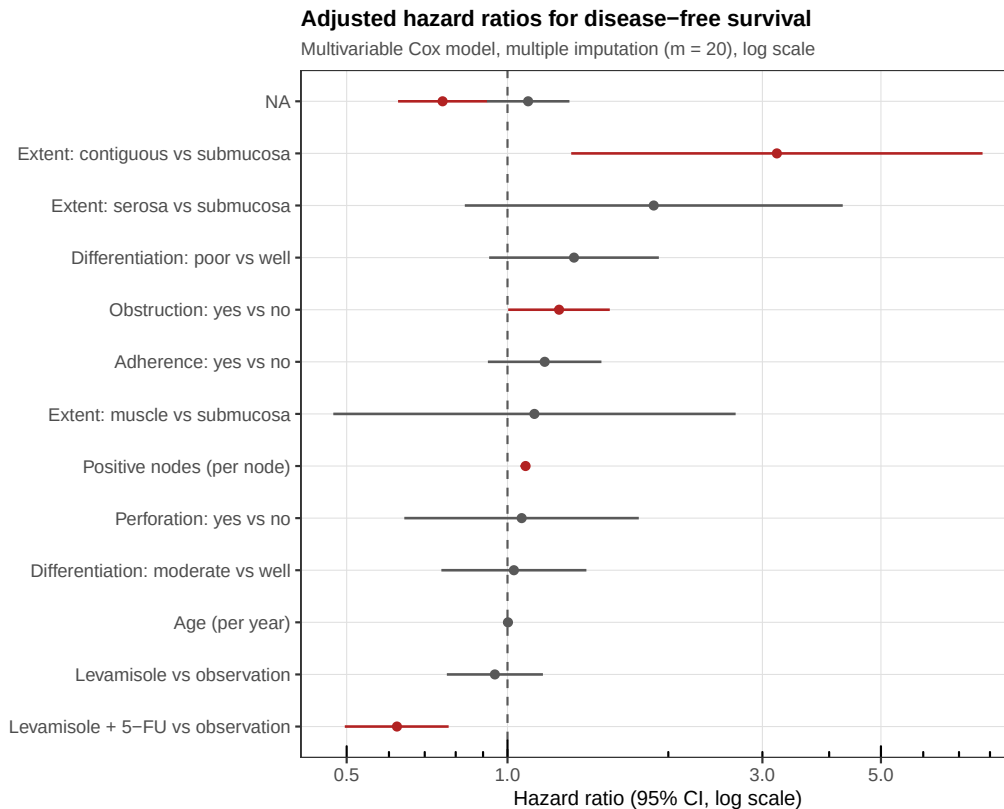


Figure 4: Adjusted hazard ratios for disease-free survival from the multiply imputed Cox model (log scale). Red indicates estimates whose 95% confidence interval excludes 1.

4.3 Absolute benefit

Five-year Kaplan–Meier DFS probabilities were 42.4% (95% CI 36.9–47.8) under observation, 44.2% (38.6–49.6) under levamisole, and 59.2% (53.4–64.5) under levamisole plus 5-FU; the corresponding overall-survival probabilities were 52.6%, 53.5%, and 63.4%. The bootstrap 95% CI for the five-year DFS risk difference of levamisole plus 5-FU against observation is -16.7 percentage points (-24.2 to -9.4), corresponding to a number needed to treat of six patients to prevent one additional DFS event at five years. Levamisole alone shows no detectable absolute benefit (-1.8 points, -9.3 to $+5.7$). These absolute magnitudes frame the trial’s clinical importance in a way the hazard ratio alone cannot.

4.4 Proportional-hazards diagnostics

The scaled Schoenfeld-residual analysis flags two covariates: tumour differentiation ($p < 0.001$) and obstruction ($p = 0.008$), with a significant global test ($\chi^2 = 42.2$, 14 df, $p < 0.001$). Reassuringly, the treatment indicator itself shows no evidence of a time-varying effect ($p = 0.51$), so the primary hazard-ratio summary is defensible. Figure 5 shows the residual patterns; the obstruction effect visibly weakens after year two, a common phenomenon for early surgical complications whose prognostic impact fades. In addition, replacing linear age and nodes with 4-degree-of-freedom penalized splines significantly improves the log-likelihood (likelihood-ratio $p < 0.001$), indicating threshold-like dose effects in the nodal count that the linear term smooths over; the treatment estimate is unaffected. Both diagnostics motivate the competing-risk sensitivity analysis and justify keeping the primary estimand simple while reporting these caveats.

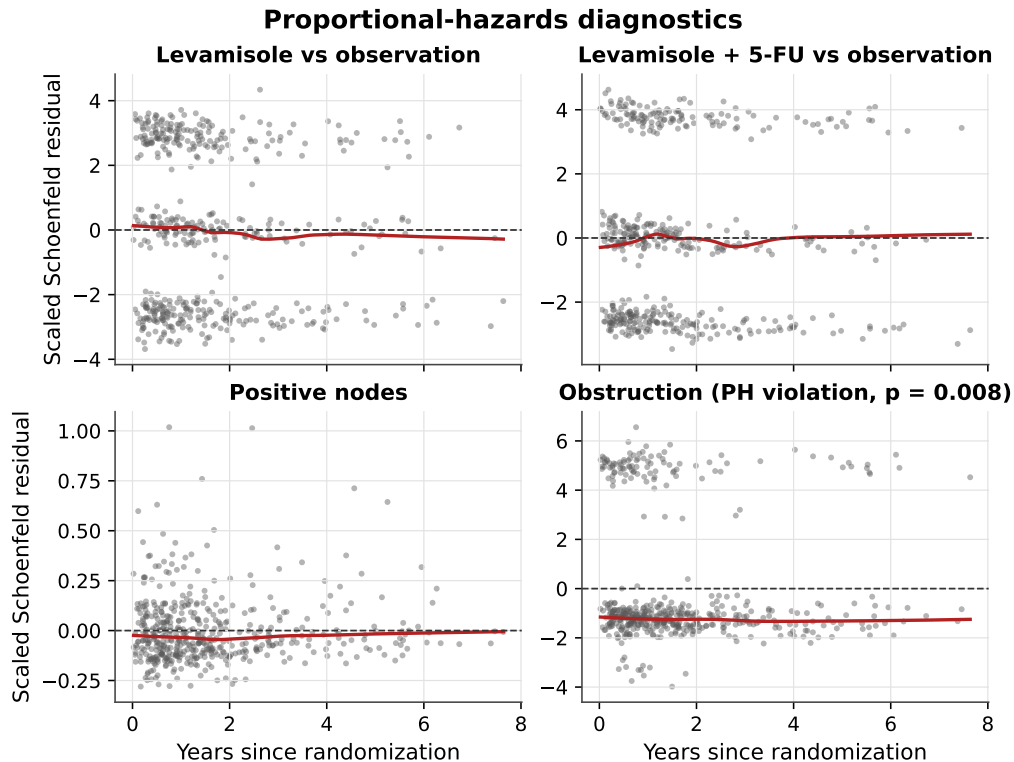


Figure 5: Proportional-hazards diagnostics: scaled Schoenfeld residuals with loess smooths for treatment, positive nodes, and obstruction (which exhibits a significant violation, $p = 0.008$).

4.5 Competing-risk sensitivity analysis

Thirty-eight patients died without a recorded recurrence, and treating these deaths as censoring—as the published coding does—formally assumes they carry no recurrence-risk information. The competing-risk view shows the practical consequence (Figure 6): the Aalen–Johansen cumulative incidence of recurrence sits below the naive one-minus-Kaplan–Meier curve for every arm, with the gap reaching about one percentage point at eight years in the observation arm. The Fine–Gray subdistribution model, adjusted for the same covariates, estimates the levamisole-plus-5-FU effect on the recurrence subdistribution hazard at 0.596 (95% CI 0.469–0.756)—virtually identical to the cause-specific reading of 0.623. The conclusion of a substantial treatment benefit is therefore robust to the endpoint-definition choice, and the analysis demonstrates the magnitude of the naive estimator’s upward bias in this dataset.

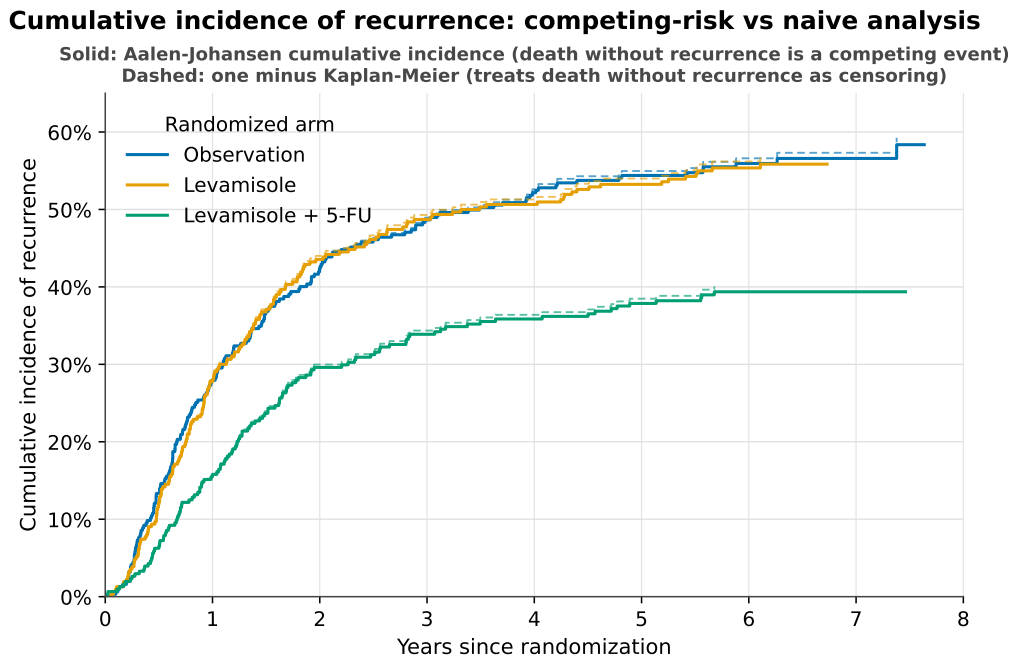


Figure 6: Cumulative incidence of recurrence: Aalen–Johansen estimates treating death without recurrence as a competing event (solid) versus the naive one-minus-Kaplan–Meier estimator that censors such deaths (dashed).

4.6 Restricted mean survival time

The RMST analysis finally restates the benefit in time units without any proportional-hazards assumption (Table 2, Figure 7). Within a five-year horizon, patients randomized to levamisole plus 5-FU can expect 3.57 disease-free years on average versus 2.94 under observation—a gain of 0.63 years (95% CI 0.33 to 0.92), roughly seven and a half months, with an RMST ratio of 1.21 (1.11 to 1.33). The gain widens with the horizon, reaching 1.13 years (0.62 to 1.65) by eight years as the curves continue to separate. The covariate-adjusted pseudo-observation regression yields 0.57 years (0.28 to 0.85), closely agreeing with the unadjusted estimate—as it must, because the RMST difference, unlike the hazard ratio, is collapsible: covariate adjustment changes precision, not the estimand. Within the obstruction stratum—the covariate with the strongest proportional-hazards violation ($p = 0.008$, Figure 5)—the five-year RMST difference remains substantial (0.60 years, 0.27 to 0.94, without obstruction; 0.83 years, 0.11 to 1.55, with obstruction), confirming that the treatment conclusion does not rest on the proportional-hazards assumption.

Table 2: Restricted mean survival time for the primary comparison (Lev+5FU vs. observation, DFS endpoint). Difference = disease-free years gained; percentile bootstrap CIs (2,000 resamples).

Horizon τ	RMST Obs (y)	RMST Lev+5FU (y)	Difference (95% CI)	Ratio (95% CI)
3	2.03	2.34	0.31 (0.14–0.47)	1.15 (1.07–1.24)
4	2.50	2.96	0.46 (0.23–0.68)	1.18 (1.09–1.29)
5	2.94	3.57	0.63 (0.33–0.92)	1.21 (1.11–1.33)
6	3.36	4.14	0.79 (0.42–1.15)	1.23 (1.12–1.37)
7	3.75	4.70	0.95 (0.51–1.39)	1.25 (1.13–1.40)
8	4.12	5.24	1.13 (0.62–1.65)	1.27 (1.14–1.43)

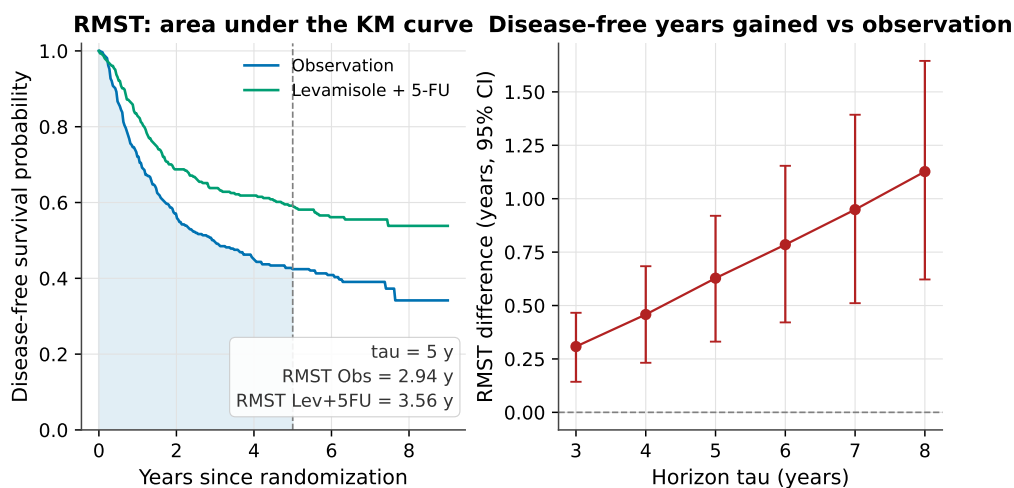


Figure 7: Restricted mean survival time analysis. Left: the RMST is the shaded area under the Kaplan–Meier curve up to the horizon $\tau = 5$ years (shown for the observation arm). Right: RMST difference (disease-free years gained with levamisole plus 5-FU) with 95% bootstrap confidence intervals as a function of the horizon.

5 Results II: Subgroups, Trial Redesign, and Re-enacted Monitoring

5.1 Subgroup analyses

Figure 8 presents the seven pre-specified subgroup analyses for the primary contrast. The hazard ratio is remarkably stable across subgroups—every point estimate lies between 0.46 and 0.81 and every confidence interval overlaps the overall estimate of 0.62. Only one nominal interaction reaches $p < 0.05$: sex (interaction $p = 0.043$), with a substantially larger effect in men (HR 0.46, 95% CI 0.33–0.64) than in women (HR 0.79, 95% CI 0.58–1.07). Against the Bonferroni-adjusted threshold of 0.0071 appropriate for seven tests, this signal does not survive, and no biological mechanism has been established for sex-specific efficacy of 5-FU-based regimens in this setting. The honest reading is *consistent relative effect across all examined subgroups*, a textbook illustration of why subgroup forest plots without multiplicity-aware interpretation manufacture false discoveries.

Subgroup analyses: disease-free survival, Lev+5-FU vs observation

Unadjusted Cox models within subgroups; interaction tests adjusted for nodal count.
Right: hazard ratios with 95% CIs and interaction p -values.

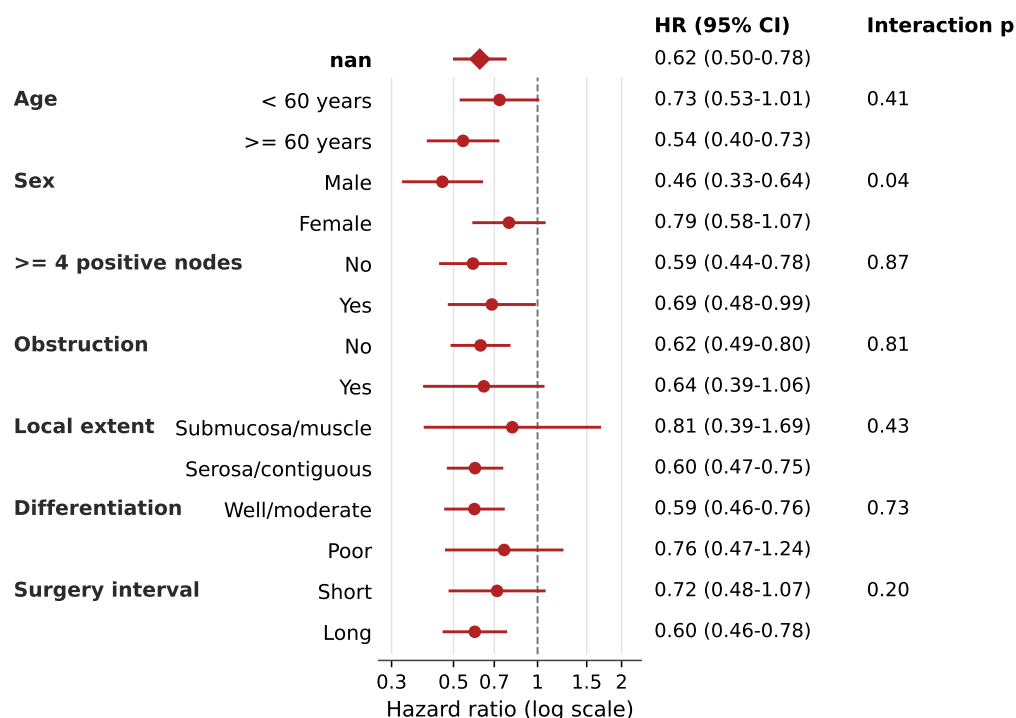


Figure 8: Pre-specified subgroup analyses for disease-free survival, levamisole plus 5-FU versus observation. Right column: hazard ratios with 95% CIs and interaction-test p -values (adjusted for nodal count).

5.2 The redesigned trial

Table 3 summarizes the fixed design and the group-sequential redesign. For the target hazard ratio of 0.65 at one-sided $\alpha = 0.025$ with 90% power, the fixed design requires 226 DFS events (approximately 322 patients under the assumed accrual and follow-up). The three-look sequential design inflates the maximum to 235 events but compensates with expected savings whenever the effect is real or absent. The O’Brien–Fleming efficacy boundary is naturally conservative early ($Z = 2.963$, one-sided $p = 0.0015$ at the first interim) and relaxes toward 2.014 at the final

analysis; the non-binding futility boundary starts at $Z = 0.079$, meaning a first look with essentially no observed trend would trigger the futility signal.

Table 3: The redesigned group-sequential trial: boundaries and operating characteristics from `gsDesign`. Efficacy uses O’Brien–Fleming alpha-spending; futility uses a non-binding Hwang–Shih–DeCani beta-spending ($\gamma = -4$).

Analysis	Events	Effic. Z	1-sided p	HR at bound	Futil. Z	HR at bound
Interim 1 (50%)	118	2.963	0.0015	0.579	0.078	0.986
Interim 2 (75%)	177	2.359	0.0091	0.701	1.006	0.859
Final (100%)	235	2.014	0.0220	0.769	2.014	0.769

Design-theoretic probabilities: $P(\text{reject} \mid \text{HR} = 0.65) = 0.900$; $P(\text{reject} \mid \text{HR} = 1) = 0.024$ (futility adhered)

Figure 9 visualizes the boundaries on the Z and one-sided p -value scales, and Figure 10 shows how power and required events vary with the target hazard ratio: the design maintains 90% power at $\text{HR} = 0.65$ with 226 events, but power falls steeply for weaker effects—at $\text{HR} = 0.80$, nearly 850 events would be required, a useful reminder of why ambitious trials with modest expected effects grow so large.

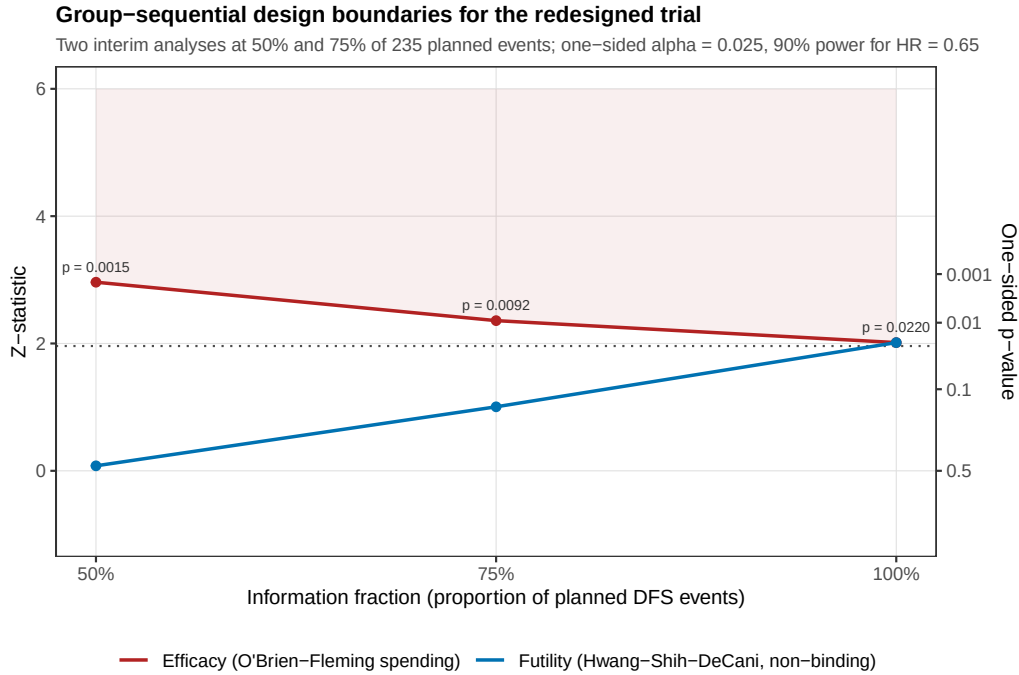


Figure 9: Group-sequential boundaries of the redesigned trial on the Z scale (left axis) and one-sided p -value scale (right axis). Red: efficacy (O’Brien–Fleming spending); blue: non-binding futility (Hwang–Shih–DeCani).

5.3 Re-enacting interim monitoring on the real data

The most striking result of this project emerges when the redesigned monitoring plan is applied to the *actual* event stream. Censoring every patient’s follow-up at the time at which the 118th DFS event was observed (information fraction 50%) and refitting the two-arm Cox model yields an interim hazard ratio of 0.52 and a Z -statistic of 3.38—*above the efficacy boundary of 2.963* (Figure 11). Under the redesign, the trial would have stopped at the first interim analysis and declared the levamisole plus 5-FU regimen superior to observation, after only 118 of the 506 events that were eventually observed.

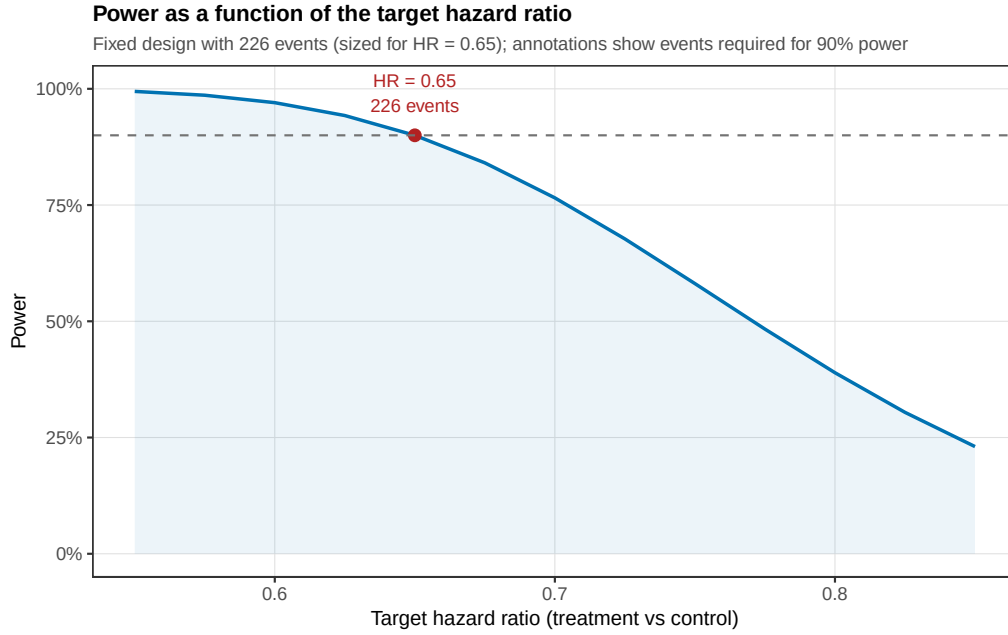


Figure 10: Power as a function of the target hazard ratio under the Schoenfeld–Freedman approximation (one-sided $\alpha = 0.025$, 1:1 allocation), annotated with the required event count at HR = 0.65.

Conditional power calculations quantify how decisive this interim result is: the probability of eventually crossing the final boundary is 70.6% even under the null, 99.8% under the design alternative, and essentially 100% if the observed interim trend persists. A cross-check with `gsDesign`’s own conditional-power machinery (`gsCP`) returns somewhat higher values (0.834 and 0.999 under the null and alternative), because it accumulates the crossing probabilities at *both* remaining analyses with boundaries re-optimized after the interim, whereas the hand-rolled calculation uses a single-look normal approximation to the final analysis only. The contrast with the Bayesian view is instructive: sequential posterior probabilities under the skeptical prior (Section 7.2) would have signalled benefit even earlier, at the 25% information look.

Two interpretive cautions apply. First, the historical trial was not designed with this monitoring plan, so the re-enactment is a counterfactual exercise, not a critique of the original conduct; its message is about how much evidence was already present early in the event stream. Second, the interim hazard ratio of 0.52 is more extreme than the final estimate of 0.62, a reminder that early random highs in Z are partly regression artefacts; the design’s conservative early boundary exists precisely to price this randomness in.

6 Results III: Simulation Studies

6.1 Study A: covariate adjustment buys power

Table 4 and Figure 12 report 1,000 simulated trials per scenario at $n = 322$ and target hazard ratio 0.72. With no prognostic covariates (scaling 0), adjustment neither helps nor hurts, as theory predicts: power is 71.6% unadjusted versus 70.4% adjusted, and the type-I error is 2.5% and 2.9% respectively (2.2% and 3.2% at full strength)—all within Monte Carlo error of the nominal 2.5%. As prognostic strength grows to the trial-calibrated value (SD of the prognostic index 0.416 on the log-hazard scale), adjusted power rises to 80.9% against 72.6% unadjusted—an efficiency gain equivalent to roughly 70 additional events (about a 23% larger event count), obtained at zero cost in bias or type-I error. This is the quantitative case behind current regulatory encouragement of pre-specified covariate adjustment [16, 15].

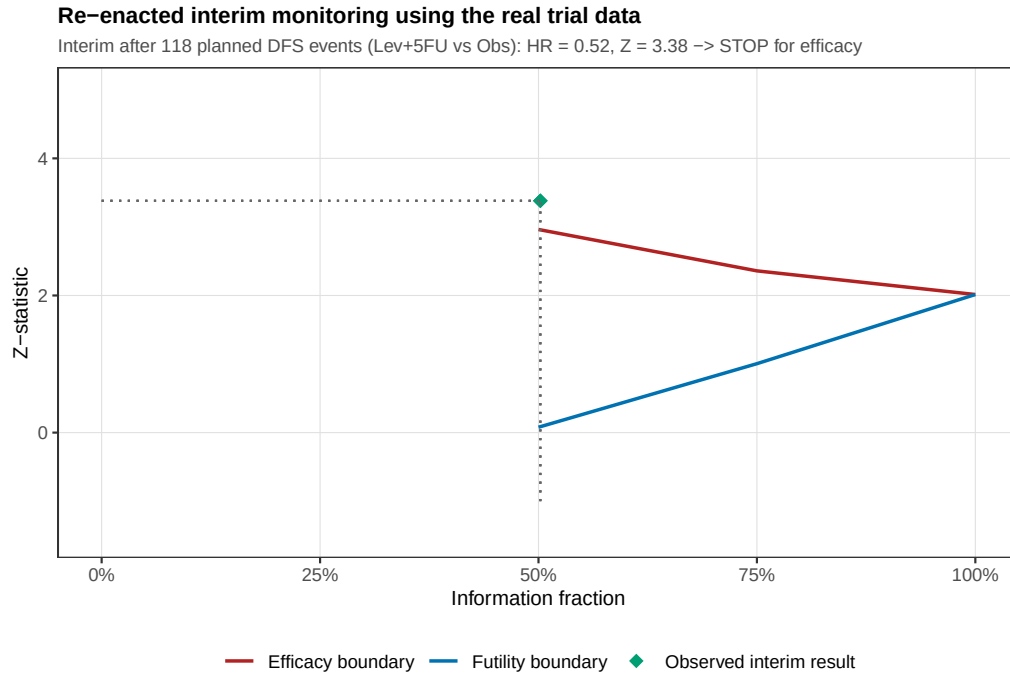


Figure 11: Re-enacted interim monitoring using the real trial data. The observed interim result after 118 events (HR 0.52, $Z = 3.38$, green marker) crosses the O’Brien–Fleming efficacy boundary (2.96), so the redesigned trial would have stopped for efficacy at the first interim analysis.

Table 4: Simulation Study A: covariate-adjustment efficiency (1,000 replicates per scenario, $n = 322$, one-sided $\alpha = 0.025$). Power at HR = 0.72; type-I error at HR = 1. “Bias” contrasts each estimator with the conditional log-HR used to generate the data.

Prognostic strength	HR	Power (unadj.)	Power (adj.)	Bias (unadj.)	Bias (adj.)
None (SD 0.000)	1.00	0.025	0.029	−0.006	−0.006
Moderate (SD 0.208)	1.00	0.025	0.025	+0.004	+0.005
Full (SD 0.416)	1.00	0.022	0.032	+0.000	−0.001
None (SD 0.000)	0.72	0.716	0.704	+0.000	−0.009
Moderate (SD 0.208)	0.72	0.725	0.766	+0.012	−0.006
Full (SD 0.416)	0.72	0.726	0.809	+0.033	−0.010

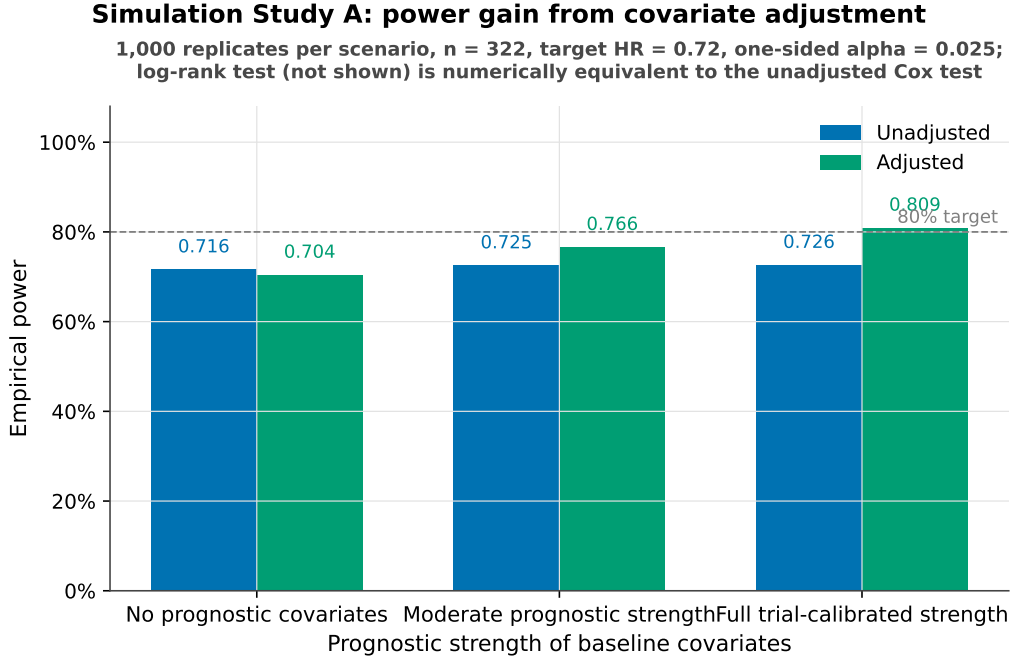


Figure 12: Simulation Study A: empirical power of the unadjusted versus covariate-adjusted Cox test as a function of prognostic strength (target HR 0.72, $n = 322$, 1,000 replicates per scenario).

The bias column also teaches a subtler lesson: at full prognostic strength, the unadjusted estimator averages $+0.033$ on the log scale relative to the data-generating conditional hazard ratio—that is, an unadjusted marginal HR of about 0.745 when the conditional truth is 0.72. This attenuation toward the null is the hazard ratio’s non-collapsibility [14]: unlike a linear-model coefficient or a risk difference, the marginal HR changes when strong prognostic covariates are removed from the model even without any confounding. Consequently, “the unadjusted and adjusted HRs estimate different quantities” is not a pedantic caveat but a measurable effect, and it explains why adjusted and unadjusted estimates in randomized trials need not coincide (and why the near-coincidence of 0.623 and 0.621 in our real data reflects the modest prognostic strength of the adjustment set rather than a mathematical necessity).

6.2 Study B: operating characteristics of the sequential design

Simulating the full monitoring procedure (3,000 trials per scenario, event-triggered looks with calendar-time censoring) validates the design and quantifies the value of the futility boundary (Table 5, Figure 13). Under the alternative (HR 0.65), the design rejects in 82.9% of trials—close to the design-theoretic 90%, with the shortfall attributable to the discreteness of event-triggered looks and the normal approximation in the boundary crossing at the interim snapshots. Stopping is dominated by the second interim (42.2%) rather than the first (18.6%), and the expected event count is 183.4 versus 235 at completion: the design saves, on average, 51.6 events (roughly 22%) when the drug works. Under the null with futility adhered to, the type-I error is 2.4% and the trial stops for futility at the first interim in 54.9% of cases—the aggressive Hwang–Shih–DeCani spending doing exactly what it was designed to do. Ignoring futility (the non-binding option) inflates the type-I error only trivially, to 2.7%, confirming the design’s stated robustness.

6.3 Study C: interval estimation under non-proportional hazards

When the true effect is time-varying (HR 0.45 in year one, weakening to 0.80 thereafter) and the analysis fits a single proportional-hazards Cox model, the estimator targets a weighted average of the two regimes (simulation estimand: $\log\text{-HR} = -0.407$, i.e. a pooled HR of about

Table 5: Simulation Study B: simulated operating characteristics of the group-sequential design (3,000 trials per scenario).

Scenario	Eff. IA1	Eff. IA2	Final	Total eff.	Exp. events	Events saved
HR = 0.65 (as designed)	0.186	0.422	0.222	0.829	183.4	51.6
HR = 1, futility adhered	0.002	0.008	0.014	0.024	152.7	82.4
HR = 1, futility ignored	0.001	0.009	0.017	0.027	234.4	0.6

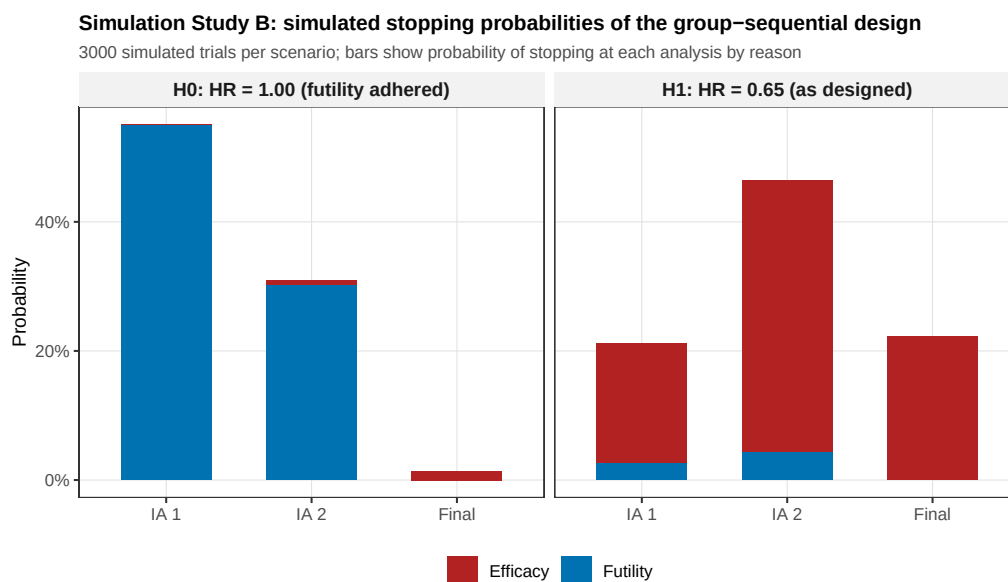


Figure 13: Simulation Study B: simulated stopping probabilities by analysis and reason under the alternative (right) and under the null with futility adherence (left).

0.666). Over 500 replicates, the Wald interval achieved 92.6% coverage of this estimand and the percentile bootstrap 90.4%—both marginally below their nominal 95% (Table 6). The bootstrap is slightly narrower (mean width 0.434 vs. 0.443) but not more accurate. The lesson is double-edged: neither standard interval collapses under moderate non-proportional hazards—the practical damage is a shifted estimand rather than drastically wrong uncertainty—yet the modest undercoverage confirms that misspecification costs are real, reinforcing the value of the diagnostics and sensitivity analyses performed in Section 4.

Table 6: Simulation Study C: 95% interval performance under a time-varying hazard ratio (500 replicates, $n = 335$; 100 bootstrap resamples per replicate).

Method	Coverage	Mean width	Mean log-HR
Cox Wald CI	0.926	0.443	−0.407
Bootstrap percentile	0.904	0.434	−0.407

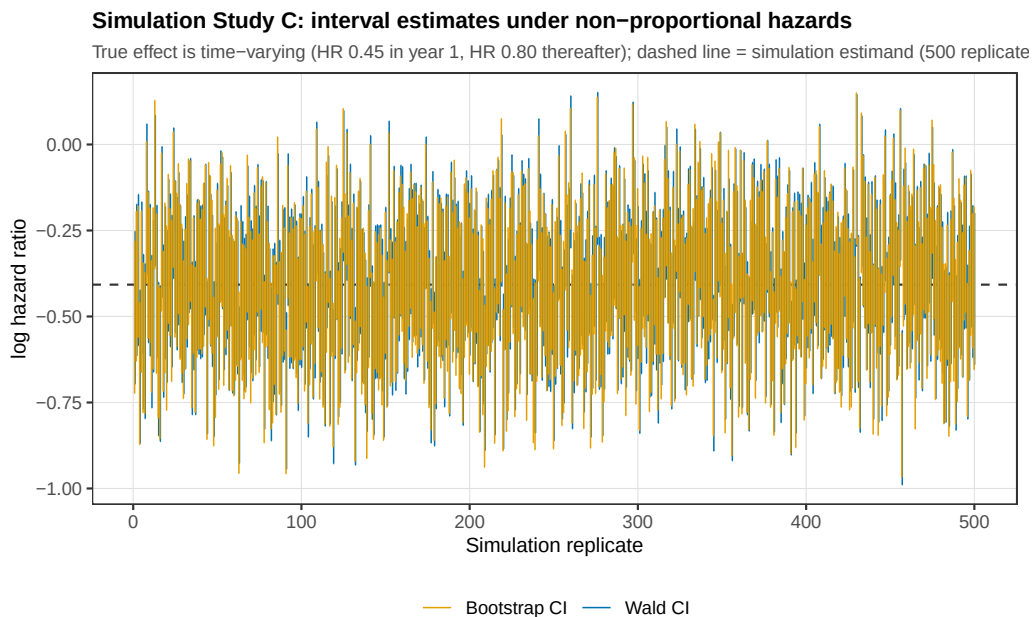


Figure 14: Simulation Study C: Wald versus bootstrap percentile intervals for the log hazard ratio across 500 simulated trials under a time-varying true effect. The dashed line marks the simulation estimand.

7 Results IV: Machine-Learning Prediction and Bayesian Inference

7.1 Cross-validated prediction performance

Table 7 and Figure 15 compare the five prognostic models over 25 cross-validation folds. The random survival forest achieves the best discrimination (cross-validated $C = 0.646$, IPCW AUC at five years 0.677) and the lowest integrated Brier score (0.217 over years 1–5), ahead of the clinical Cox model ($C = 0.634$), gradient boosting (0.633), XGBoost (0.625), and the elastic-net Cox model (0.620). The spread among models is modest—about 0.026 in the C -index—and the honest uncertainty (fold-to-fold SD ≈ 0.025) cautions against declaring a definitive winner; what the comparison really shows is that, with ten baseline covariates and ~ 300 events, flexible learners extract only a little more signal than a well-specified clinical model.

Table 7: Machine-learning prediction of the composite DFS endpoint (888 complete cases; 5×5 -fold stratified cross-validation). Treatment excluded from all predictors. AUC at five years is IPCW weighted; IBS is the integrated Brier score over years 1–5.

Model	CV C -index	CV AUC (5y)	CV IBS	Apparent C
Random survival forest	0.646	0.677	0.217	0.713
Clinical Cox (main effects)	0.634	0.652	0.222	0.651
Gradient boosting (Cox loss)	0.633	0.662	0.225	0.717
XGBoost (Cox objective)	0.625	–	–	0.741
Elastic-net Cox	0.620	0.646	0.226	0.628

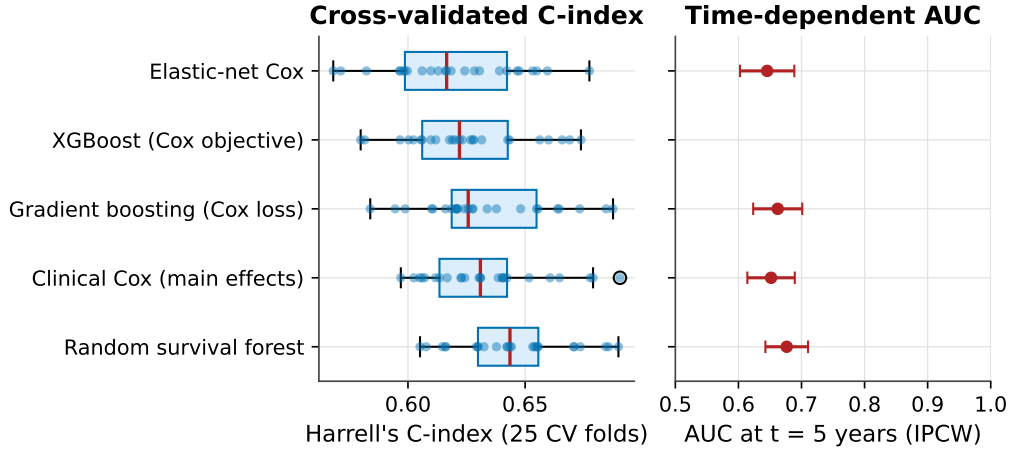


Figure 15: Cross-validated discrimination by model. Left: distribution of the C -index across 25 folds. Right: IPCW AUC at five years with cross-validation standard errors.

The apparent-versus-cross-validated contrast in Table 7 quantifies overfitting: XGBoost’s apparent C of 0.741 drops to 0.625 under honest validation, a 0.12 optimism gap, versus a 0.07 gap for the random forest and essentially none for the clinical Cox model. This gap is the single most important column in the table—it is the difference between the performance one would report from a naive in-sample fit and the performance a new patient would actually experience.

Calibration of the winning model (Figure 16, right panel) is satisfactory across the risk spectrum: within deciles of predicted five-year risk, the mean predicted risk tracks the Kaplan–Meier observed risk within 6.5 percentage points, with a slight over-prediction in the lowest decile (27.6% predicted vs. 21.4% observed). The five-year time-ROC achieves an AUC of 0.67 (Figure 16, left panel). Permutation importance (Figure 17) is dominated by the positive lymph-node count (mean C -index drop 0.117 per permutation, an order of magnitude above every other covariate), followed at a distance by advanced local extent, female sex, and a long surgery-to-registration interval—a ranking that mirrors the adjusted Cox effects of Figure 4 and reinforces the clinical coherence of the learned model.

7.2 Bayesian re-analysis: priors and posteriors

The Weibull model converged cleanly under all three prior scenarios (two chains, target acceptance 0.9; $\hat{R} \approx 1.00$ throughout). Table 8 and Figure 18 display the central result: the posterior hazard ratio is essentially insensitive to the prior, ranging only from 0.591 (weakly-informative) to 0.610 (skeptical). Under the skeptical prior—which places 95% of its mass between hazard ratios of about 0.44 and 2.28 and only 5% below 0.5—the posterior mean is 0.610 with 95% credible interval 0.485–0.757, $\Pr(\text{HR} < 1) = 1.000$ to three decimals, and $\Pr(\text{HR} < 0.8) = 0.994$.

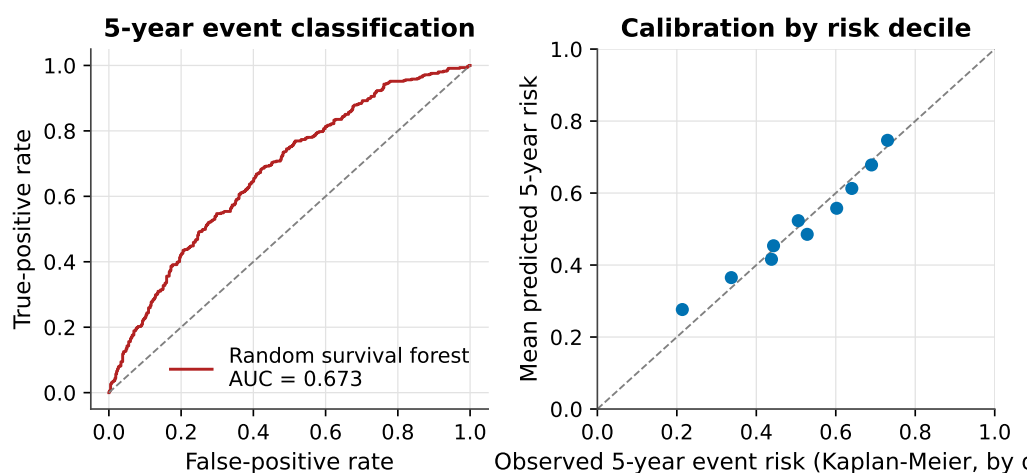


Figure 16: Discrimination and calibration of the random survival forest. Left: ROC curve for five-year event classification (AUC = 0.673). Right: predicted versus Kaplan-Meier observed five-year event risk by decile of predicted risk.

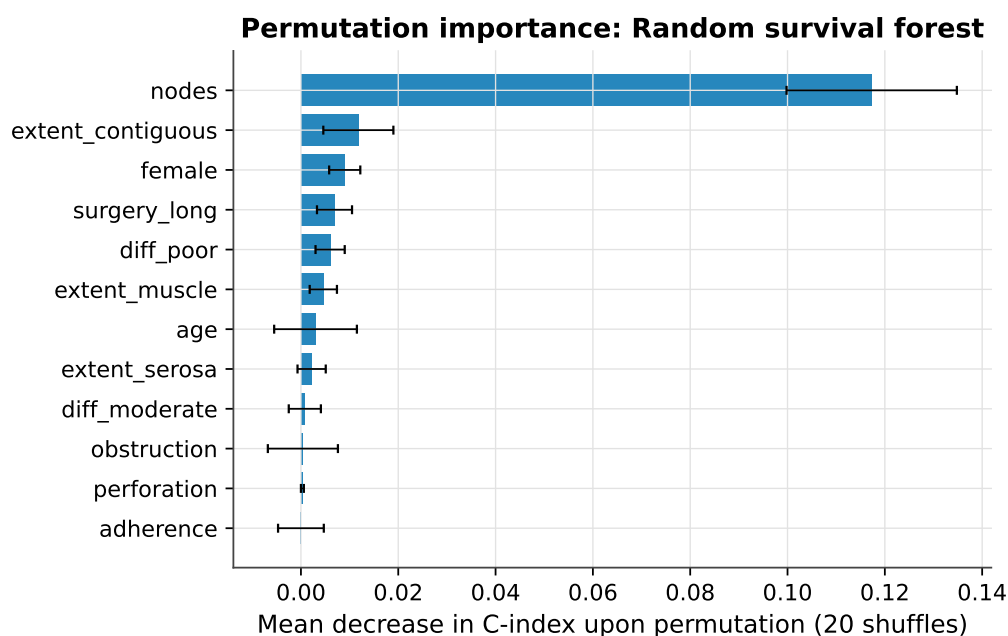


Figure 17: Permutation importance for the random survival forest (mean decrease in C -index over 20 shuffles per covariate, with standard deviations).

The data are simply strong enough to overwhelm any of the three priors, which is itself an informative finding: conclusions of substance do not require prior consensus.

Table 8: Bayesian Weibull analysis of the primary contrast under three prior specifications (2 chains; 4,000 posterior draws each), with the frequentist adjusted estimate for comparison.

Prior	Mean (log-HR)	Post. HR	95% CrI	Pr(HR<1)	Pr(HR<0.8)
Skeptical $\mathcal{N}(0, 0.421^2)$	0	0.610	0.485–0.757	1.000	0.994
Weakly-inf. $\mathcal{N}(0, 1)$	0	0.591	0.466–0.738	1.000	0.997
Enthusiastic $\mathcal{N}(\log 0.65, 0.421^2)$	−0.431	0.592	0.474–0.734	1.000	0.997
Frequentist (adj. Cox, MI) [ref.]	—	0.621	0.496–0.776	—	—

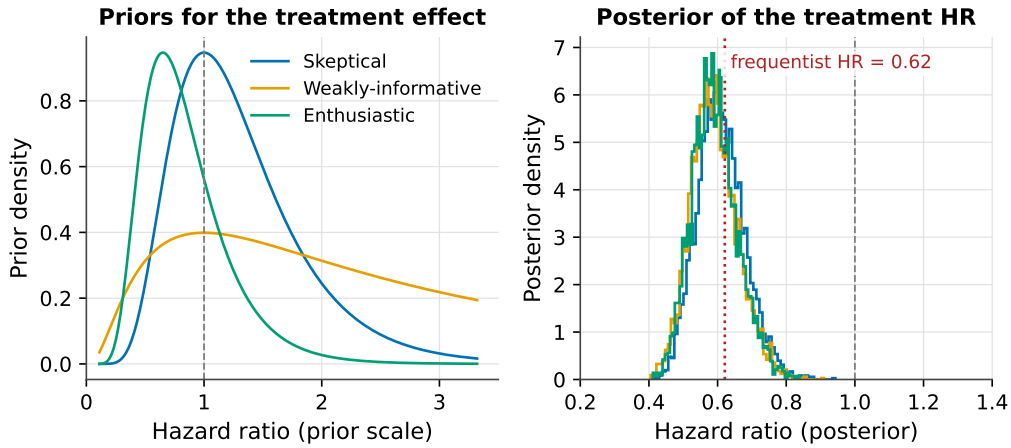


Figure 18: Prior sensitivity of the Bayesian treatment-effect estimate. Left: the three prior densities on the hazard-ratio scale. Right: the corresponding posterior densities, with the frequentist point estimate (dotted) for reference.

7.3 Bayesian sequential monitoring

Figure 19 replays evidence accumulation under the skeptical prior at four event-driven looks. By the 25% look (79 events), the posterior mean hazard ratio is already 0.601 with $\Pr(\text{HR} < 1) = 0.998$ and a credible interval excluding 1 (upper bound 0.88); from the 50% look (158 events) onward the probability is essentially 1. Under the common Bayesian monitoring rule of stopping when $\Pr(\text{HR} < 1)$ exceeds 0.975, this trial would have stopped at the *first quarter* of the evidence—even earlier than the frequentist group-sequential re-enactment of Section 5, which required 50% of the information and a much larger Z . The contrast is partly definitional (the Bayesian posterior probability does not pay a multiplicity price for looks) and partly prior-driven (the skeptical prior pulls toward the null early on, making its rapid crossing more impressive); both mechanisms deserve emphasis when Bayesian designs are proposed as replacements for alpha-spending.

7.4 Model criticism

The posterior predictive check (Figure 20) replicates the control arm’s Kaplan–Meier curve from 100 posterior draws: the observed curve lies comfortably within the 95% posterior predictive band throughout follow-up, with no systematic deviation in the tail where a poorly specified baseline hazard would typically betray itself. Approximate leave-one-out cross-validation strongly prefers the Weibull baseline over the exponential special case (ELPD difference -18.0 , all stacking weight on the Weibull), and the posterior shape parameter $\alpha = 0.765$ (95% CrI 0.695–0.837) confirms a genuinely decreasing baseline hazard—patients who survive the first two years have

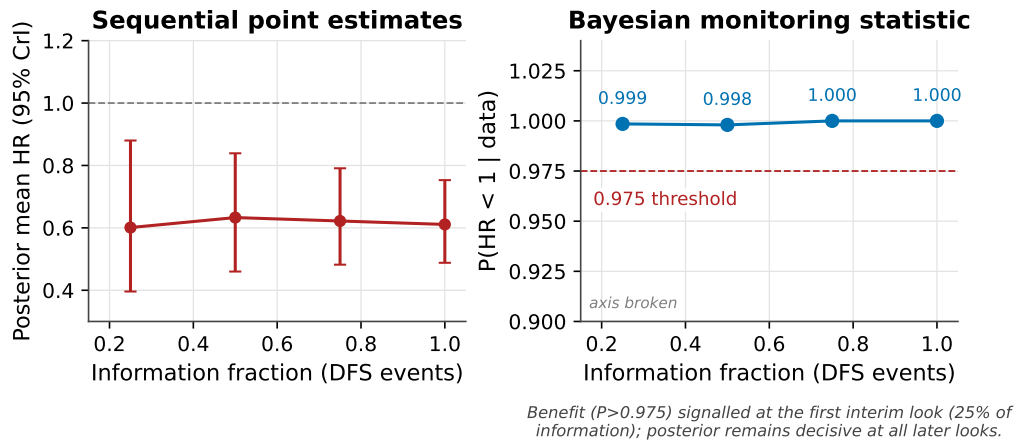


Figure 19: Bayesian sequential monitoring under the skeptical prior. Left: posterior mean hazard ratio with 95% credible intervals at each look. Right: the monitoring statistic $\Pr(HR < 1)$ against the 0.975 stopping threshold. The right panel’s y axis is broken at 0.9 so the small variation between looks (0.998, 0.998, 1.000, 1.000) is visible; the line is flat near the top because the posterior probability of benefit is already overwhelming at the first interim look.

progressively lower event rates, exactly as clinical experience suggests and as the Kaplan–Meier shape in Figure 2 already indicates.

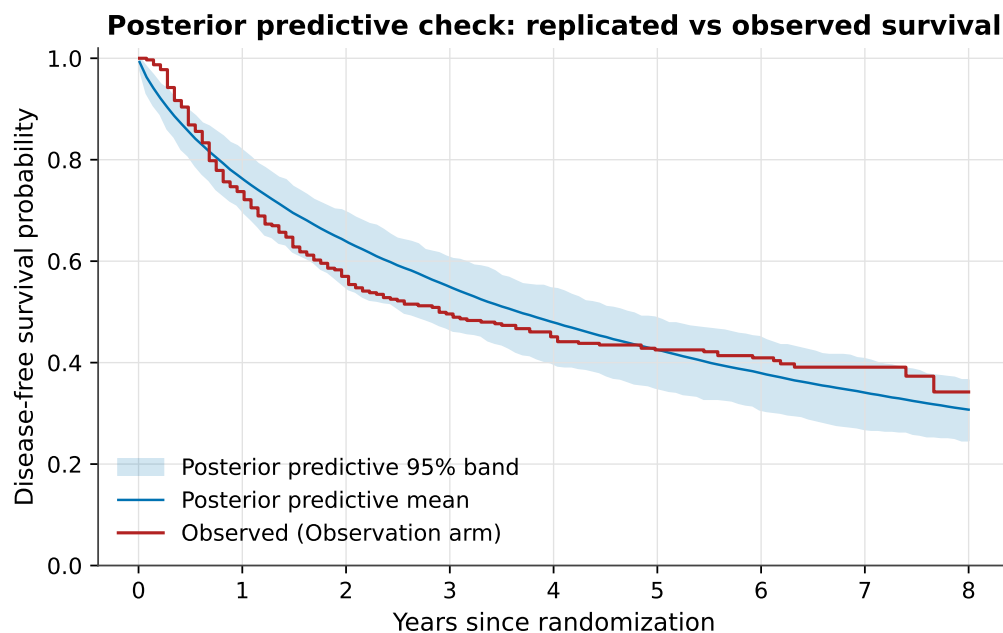


Figure 20: Posterior predictive check for the control arm: replicated Kaplan–Meier curves from 100 posterior draws versus the observed curve.

8 Results V: Heterogeneous Treatment Effects

8.1 Average effects on the five-year absolute risk

Table 9 reports the average treatment effect of levamisole plus 5-FU versus observation on the five-year DFS event risk, estimated three ways on the jackknife pseudo-observations ($n = 594$ complete cases). The three estimators agree closely: the unadjusted Kaplan–Meier difference is -16.8 percentage points (bootstrap 95% CI -24.5 to -9.1), cross-fitted G-computation gives

−15.3 points, and the doubly robust AIPW estimator gives −16.2 points with influence-function standard error 3.9 (95% CI −23.9 to −8.5). The consistency across a design-free estimator, an outcome-model-based estimator, and a doubly robust estimator is exactly what one hopes to see in a randomized comparison, and the magnitudes match the ITT analysis of Section 4.

Table 9: Average treatment effect on the five-year DFS event risk (negative = risk reduction with levamisole plus 5-FU).

Estimator	ATE (percentage points)	SE	95% CI
Kaplan–Meier difference (bootstrap)	−16.8	4.0	−24.5 to −9.1
G-computation (cross-fitted, pseudo-obs)	−15.3	0.5	−16.2 to −14.4
AIPW (cross-fitted, doubly robust)	− 16.2	3.9	−23.9 to −8.5

8.2 Conditional effects: where the benefit lives

The cross-fitted T-learner decomposes this average into patient-level predicted benefits (Figure 21). The distribution is wide: predicted five-year risk reductions range from negligible to over 40 percentage points, with quartile means of 28.9, 18.6, 12.6, and 1.1 points. Crucially, the predicted benefit is strongly tied to baseline risk (correlation 0.648 between control-arm predicted risk and predicted benefit; right panel of Figure 21): the patients predicted to benefit most are those whose disease is most aggressive—high nodal burden, advanced local extent—while low-risk patients have little room for absolute improvement even under a proportionally constant treatment effect.

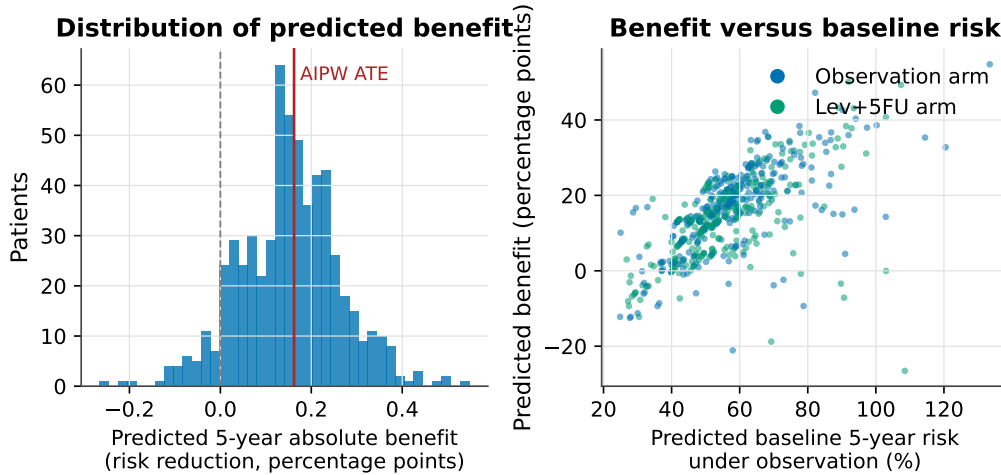


Figure 21: Predicted absolute benefit of levamisole plus 5-FU at five years. Left: distribution across patients (AIPW average marked). Right: predicted benefit versus predicted baseline risk, coloured by randomized arm.

Table 10 sharpens the picture with observed outcomes: within the highest predicted-benefit quartile (mean baseline risk 71.0%), the observed five-year risk difference is −12.8 points; within the lowest quartile (baseline risk 46.6%), it is −19.3 points—statistically compatible with the same relative effect everywhere. The apparent paradox (large predicted but modest observed benefit in Q1, the reverse in Q4) reflects both estimation noise in quartile-specific contrasts (each based on ~64–98 events) and the mechanical relationship between baseline risk and achievable absolute benefit. The Kaplan–Meier curves within the extreme quartiles (Figure 22) visualize the trade-off: in the high-benefit quartile both arms fall steeply but the curves separate; in the low-benefit quartile both arms do comparatively well.

Table 10: Predicted-benefit quartiles from the cross-fitted T-learner. “Predicted benefit” is the model-implied five-year risk reduction; “observed risk difference” is the corresponding pseudo-observation contrast by arm within quartile.

Quartile	<i>n</i>	Pred. benefit (pts)	Baseline risk	Observed diff (pts)	Events
Q1 (largest predicted benefit)	149	28.9	71.0%	−12.8	98
Q2	148	18.6	58.2%	−16.1	79
Q3	148	12.6	52.0%	−11.8	72
Q4 (smallest)	149	1.1	46.6%	−19.3	64

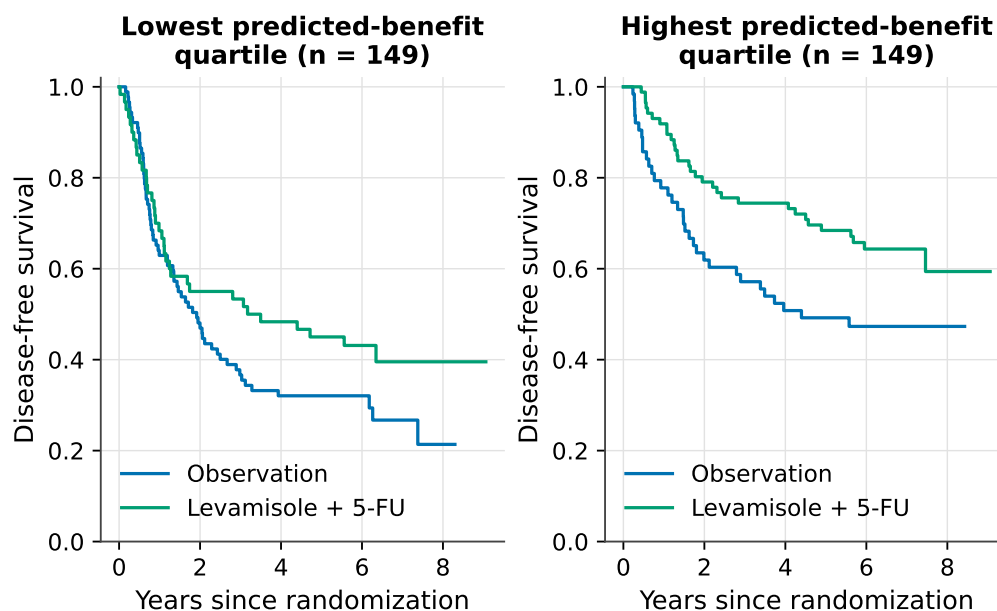


Figure 22: Kaplan–Meier disease-free survival within the lowest and highest predicted-benefit quartiles.

The synthesis is the section’s key message: in this trial, heterogeneity in *absolute* benefit is real and large, but it is driven almost entirely by heterogeneity in *baseline risk* rather than by qualitative effect modification—consistent with the null interaction tests of Section 5. For clinical decision-making, this is the most useful kind of heterogeneity: it means a single relative effect, applied to an accurate prognostic model, suffices to identify who gains most in absolute terms, with no need to invoke distinct treatment-response biology. All such CATE statements remain exploratory in a single trial of this size, and the honest framing is that the analysis demonstrates the *methods* and generates hypotheses for external validation.

9 Discussion and Conclusions

9.1 Principal findings

Four independent analytical paradigms, aimed at one 1980s dataset, converge on one conclusion and enrich it from different angles. The conclusion: levamisole plus 5-FU after resection of stage B/C colon carcinoma delivers a large, robust treatment benefit on disease-free survival—hazard ratio 0.62 adjusted, 0.59–0.61 Bayesian, 0.60 under competing risks—corresponding to a five-year absolute gain of 16.8 percentage points, a number needed to treat of six, and 0.63 disease-free years gained in restricted mean survival time. The enrichment: the effect is stable across seven pre-specified subgroups with no credible interaction; a modern group-sequential design would likely have stopped the trial for efficacy at its first interim analysis (HR 0.52, $Z = 3.38$ against a boundary of 2.96), and Bayesian monitoring would have signalled benefit at a quarter of the evidence; covariate adjustment would have bought real power at no inferential cost; and the largest absolute benefits accrue to the highest-risk patients through the arithmetic of baseline risk rather than differential response.

The re-enacted monitoring result deserves particular emphasis because it connects historical data to contemporary trial ethics. Fifty percent of the 506 DFS events in this dataset contain essentially all the evidence needed for the superiority conclusion; the remaining half of the patients’ events—years of follow-up and hundreds of recurrences—added precision but not decisions. Translated to the 1980s context in which the 625 patients randomized to observation or levamisole alone in this very trial continued on an ineffective strategy, an early stop would have redirected patients to the active regimen years sooner. The counterfactual is stylized (the historical investigators lacked today’s data-management speed and regulatory framing), but it illustrates with unusual concreteness why alpha-spending and beta-spending designs became the default.

9.2 Methodological lessons

Several results function as transferable teaching points. **Non-collapsibility**, usually a paragraph in a textbook, appears here as a measured quantity: with trial-calibrated prognostic strength, the unadjusted Cox estimator targets a marginal hazard ratio attenuated by 0.033 on the log scale, so unadjusted and adjusted estimates are *different estimands* even in a perfectly randomized trial (Simulation Study A). **Covariate adjustment** raises power from 72.6% to 80.9% in exactly the scenario regulators care about—modest effects, realistic sample sizes—which is the quantitative argument behind the FDA/ICH position that adjustment should be pre-specified and reported even when unadjusted results match [16, 15]. **Competing risks** change the estimand, not necessarily the conclusion: the naive one-minus-Kaplan–Meier overstates cumulative recurrence by about one percentage point at eight years, yet the Fine–Gray and cause-specific readings agree on the treatment effect (Section 4). **Honest validation** matters more than model sophistication: the apparent-to-cross-validated optimism gap reaches 0.12 in the C -index for the most aggressive learner, while a clinical Cox model loses nothing (Ta-

ble 7)—a reminder that in small-to-moderate clinical datasets, disciplined regression frequently matches ensembles. **Bayesian monitoring** demonstrates both its appeal (decisions at 25% of the information) and its governance question (posterior probabilities do not self-correct for repeated looks, so their use in confirmatory settings requires either calibration to frequentist error rates or a genuinely decision-theoretic framing).

9.3 Strengths

The project’s strengths are its integration and its reproducibility. Every module runs from version-controlled code with fixed seeds, communicates exclusively through files, and regenerates every number and figure in this report via a single `make` invocation. Quality-control assertions are embedded in the data pipeline (record counts, within-patient covariate consistency, endpoint logic), analyses are pre-specified in structure (primary composite endpoint, adjustment set, subgroup list), and sensitivity analyses were run in both directions—towards stronger assumptions (parametric Weibull Bayesian model) and weaker ones (competing risks, splines, complete case versus imputation). The simulation studies were calibrated to the real data-generating mechanism rather than to stylized parameters, so their conclusions transfer directly to trials of this type.

9.4 Limitations

Five limitations bound the interpretation. First, this is a *re-analysis of a single historical trial* from an era of different supportive care and staging; the quantitative findings (especially absolute risks) should not be read as contemporary practice, and levamisole itself was later displaced by superior regimens. Second, the subgroup and CATE analyses are exploratory: with only 324 events in the two-arm comparison, interaction tests have limited power, the T-learner’s quartile contrasts rest on 60–100 events each, and no external validation cohort was available to check the predicted benefit ranking; the honest claim is methodological demonstration, not clinical deployment. Third, the Weibull baseline is a parametric choice; although posterior predictive checks pass and LOO prefers it over the exponential, a piecewise-exponential or semi-parametric Bayesian baseline would be a natural robustness extension, as would a time-varying treatment effect in the Bayesian model given the covariate-level PH violations detected. Fourth, the re-enacted monitoring and Bayesian sequential analyses re-use the final data to reconstruct interim states; the reconstruction censors follow-up at a common time aligned to the event count and is faithful to the ordering of the event stream, but the exercise remains a counterfactual simulation, and early-stopping estimates from such re-enactments are biased toward larger effects (as the interim HR of 0.52 versus the final 0.62 illustrates). Fifth, the prediction models use only ten baseline covariates from the original case-report forms; modern depth of data (genomics, imaging) would shift the accuracy ceiling, though the overfitting lesson of Table 7 would only sharpen.

9.5 Conclusions

This capstone set out to demonstrate what a modern biostatistical toolkit, applied with discipline to a classic randomized trial, can extract beyond the original publication—and the answer is: the same conclusion, reached faster (sequential design), quantified more honestly (imputation, competing risks, absolute effects), understood more deeply (simulation-calibrated operating characteristics, Bayesian robustness), and personalized in direction (risk-driven absolute benefit). The accompanying repository delivers the full pipeline—R and Python, 22 figures, 37 tabular outputs, and this report—reproducible with one command, in the hope that it serves both as a portfolio piece and as a teaching resource on how frequentist, Bayesian, and machine-learning perspectives can be made to interrogate the same data in complementary ways.

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